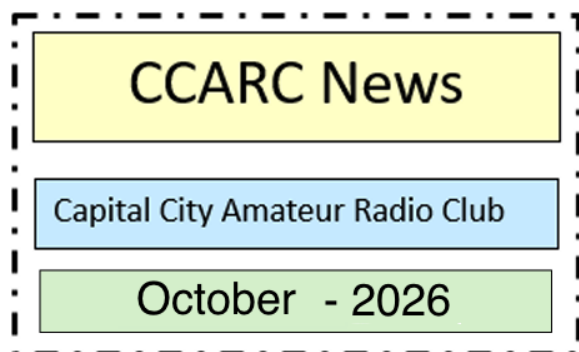




CCARC Annual Picnic

The CCARC held its annual club picnic on Saturday, 12 September. This year's event had a spectacular turnout at the Lewis and Clark County Fairgrounds. Area hams drove in to join us from as far away as Missoula, Great Falls, Anaconda, Bozeman and parts in between.

From most people's recollections, this was one of the best attended picnics in recent memory. It was great to see so many friends reconnecting, and new friendships being formed. More than a few deals were transacted at the tailgate swap meet, and the food was great too. Thank you to everyone who helped with the planning, setup, cooking, and clean up.

Upcoming Events - 2026

CCARC monthly meeting – Nov. 2, 7 p.m. at the Salvation Army, 1330 Hudson St. or the Google Meet link available on our n7rb.org/meetings-nets page

The Great Montana Shake Out - 15 Oct. 10:15 a.m. Net on the Belmont Repeater.

Just For The Helena Of It race - 18 Oct. Contact Al Simons WA1TYB if you would like to help support this event.

Coffee Net – Every Saturday at 9 a.m. on 147.22, Belmont Repeater. Eyeball @ Red Atlas at 9:30 a.m.

Trivia Net - Every Sunday, 7:30 p.m. on 147.22. John Monson AJ7MT





Belmont Tower Maintenance

After over a year of discussion, and months of planning, a faithful group of CCARC members recently assembled on top of Mount Belmont. Their mission: improve the club's repeater tower guy wires and grounding.



Our trusty N7RB repeater on Belmont has a tough, solitary life. Most of the year it endures extreme weather and

very windy conditions. All that continual exposure to the elements eventually wears things out. For some time, those symptoms of wear-and-tear made it difficult to use the repeater on very windy days. We suspected loose connections or grounding on the tower or guy wires was inducing Radio Frequency Interference (RFI). This especially negatively affected weaker stations like anyone on a HT radio. The repeater's transmitted signal was full of static and noise on windy days. We knew something had to be done.



This was actually part two of the club's recent efforts to make improvements to our repeater; putting your club dues to good use. Last summer the whole repeater was replaced with a newer model that added a few more

capabilities and improved reliability. This year we had to work on the guys. Next year - possibly adding solar power.

The crew went up on a beautiful Montanan day, Thursday (24 September). They installed new guy clamps and insulators to prevent spurious currents between the guy wires that were potentially causing static during high wind events. They also properly set the guy wire tensioning to ensure the tower stays where we want it: upright.

This maintenance effort could not have come any sooner. The crew found one guy wire significantly under tension. They also found worn bolts that definitely needed to be replaced. With any luck, our repeater tower will have many more useful years ahead of it, positioned properly: standing upright.





Reports about the repeater's performance after the work was completed have been very positive. Many people commented that the audio and signals are greatly improved.

A special thanks to the repeater maintenance crew: Brian Lee - KJ7OUF, Al Simons - WA1TYB, Bill McGuire - N7MSI, Eric Webb - KE7NLU, Stacy Webb - KK7CJV, Matt Gould - AA7MG, Keith Platts - KM7AEM, Mark Heckenlaible - W7HS, Dan Hawkins - N7SHM, and Steve Miller - KF7HJS.

SOTA Trip Report

By Rob Kingery

We recently returned from our Denver to Las Vegas trip. Although the purpose of the trip was to attend a nephew's wedding in Denver and to see Barb's

parents in Las Vegas, we did manage to complete three SOTA activations.

The first activation was at Muddy Mountain (W7Y/NW-063). This was a super convenient summit just south of Casper that had a nice BLM campground (Rim Campground) at the summit. I was able to complete my activation from our picnic table, only five or ten feet below the summit – well within the activation zone. I made twenty QSOs that evening, and another eleven the next morning, including Bob-WB7BWZ here in Montana, and a JA on 40 meters!



After Muddy Mountain and the Denver Wedding, we drove to Las Vegas where we camped just west of town. It was a warm time in Las Vegas with temperatures of up to 110 degrees. We activated Red Benchmark (W7N/CK-177) on the first morning, and found that the heat, even early in the morning made it a tough 1-pointer! Nonetheless, I did manage to get fourteen QSOs, including another one with WB7BWZ.



We made it a two-day trip back to Helena, with an overnight stay in Logan Canyon. On the way from Las Vegas to Utah, we activated Navajo Point (W7U/IR-002). This is a super easy 10-pointer that is located on top of the western side of Brian Head Ski area, just north of Cedar Breaks National Monument. It was a nice time to be there, as the temperatures were pleasant and the aspens were just beginning to turn. I made twenty QSOs, including six Summit-2-Summits, and Bob-WB7BWZ – giving Bob a clean sweep of all of the summits on my trip!



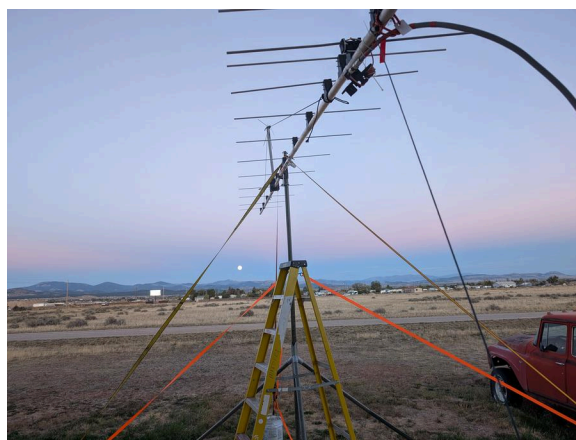
73,

Rob - AE7AP

Is the moon within reach?

By Tom Mandera

I had thought about trying Earth-Moon-Earth (EME) for some time. I picked up a 100W 2m SSB radio at one of our SK sales a few years back, and picked up a fourteen-element CP KLM antenna from eBay, but had never given it a real try.



N7SKI asked who might be able to do an EME contact that he could participate in, and I was reminded that I allegedly had the right stuff to try, but hadn't.

Some research said maybe I wasn't as close as I thought. The old Icom has too much frequency drift for modern JT65 and Q65 codecs and wasn't designed for sixty-second key downs. The fourteen-element antenna is really seven elements as far as gain, and was likely on the frustrating end of things. I decided a new antenna was appropriate and set out to build a twelve-element yagi on a twenty-four foot boom.

Then it was time to get it on the air. I set up a tripod of military camo poles and set the antenna at twelve feet after consulting various scenarios with ground gain at eight feet vs twelve feet of elevation (multiples of the four foot poles)

On Saturday evening, 9/26, I warmed up the Icom 706 MKIIG for a few hours in advance (to heat soak it and limit thermal drift), and Brett N7SKI and Devin N7HNT came by to see the action.



I used a Signalink to the laptop. The amplifier was an RF Concepts 2-170 that is a 45W in for 170W out that came from Bob W7AFM (SK)'s estate, with a computer fan strapped to the top to keep it cool since it wasn't designed for the sixty-second keydown. My ancient 35A Astron supply kept everything running.

We ran the 706 at about half power to keep radio heating down (which leads to

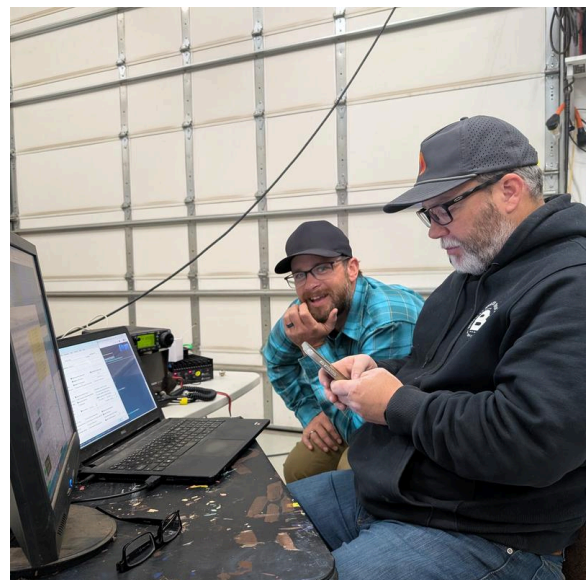
drift) and to make sure the amplifier stayed happy.

We had about an hour of moon between three degrees (lower is blocked by mountains) and twelve degrees (at which point our ground gain disappears) and managed to successfully get decoded by two "big gun" stations!

```
Tom, ur best decode 120 109 34
0143 2.8 -20 # CQ KE7VUX DN36
===== {K9MRI Joe IN EN70iu } 27Sep
01:45 0143 -28 2.8 1282 #* CQ
KE7VUX DN36 =====
{KB9TVR/4X8/KW Robin IL EM49tm }
```

K9MRI is using an eight antenna array of twenty-eight elements (XPOL, or fourteen element horizontal and fourteen vertical) antennas and an amplifier; 23 dBi gain compared to our 15.5. EIRP of 144-195kW.

KB9TVR uses 4x8 element yagis. 17.5-18.5 dBi. EIRP of 56kW to 70kW.



We thought we heard signals, but were unable to decode them at all that night. FT8 tests were successfully over on 40m to verify things should work, but JT65 on VHF was not happening.

I believe we had some Windows audio settings that were “helping” us by hindering the decodes. FT8 is more forgiving.

Inspired, we called it a night and returned on Sunday to try again. Sadly, while we may have addressed the decode issues, we found no signals to decode. EME'ers in the chat rooms were complaining about bad conditions, and the big guns we'd tried to work on Saturday were not present.

I tried again on Monday night with no better results. I did get a -30 signal report from a big gun in Russia, but I once again heard nothing in response. I have another radio to try, but ultimately the advice I kept getting that I NEEDED a low-noise-amplifier at the feed point was probably correct. None of my radios have sufficiently “quiet” receivers to decode the signals – even with a short run of high-end coax.

If I get that missing LNA, the next good time is moon-set on 11/1 during the EME contest.

Brass Knuckle Gang (BKG) Goes DX

By Devin Felix N7HNT

A tale of the first-ever DX BKG induction.

It all began with an email. A few other Montana BKG members and I received an email from Vince VE6LK stating that he would really like to be jumped into BKG. Naturally every one of us were excited at the prospect of being the first ones to get the gang across the border into Canada!

After a couple messages back and forth I remembered that I already had a trip up north to Kalispell planned to go check out a comedy show later in the month. I asked Vince if he was up for a drive to the border where we could get this done and he was all for it until I told him the date that I planned on being there. Turns out, this was actually his anniversary so we initially scrapped the idea and I figured one of the other esteemed members would have to make the epic journey northward.

Two weeks later I received another message saying he had spoken to his wife and somehow convinced her to spend their special day driving cross-country for nerd things and the quest was back on! I did a bit of Google scouting to try and figure out a spot I might be able to set up and found a

relatively new POTA location that would get me within two miles of the border. Obviously this was my only choice. How could I pass up the chance to get a new park under my belt?



When that fateful morning arrived my XYL and I were up early to find a great breakfast spot before we left town. Side note: if you are ever in Kalispell, the Sunrise Café is absolutely amazing. Then we were on our way! It was only about an hour's drive to the Kootenai Woods Ranch Wildlife Management Area (POTA reference US-12656) about two miles north of Eureka. Most of the drive up was through fairly thick fog and we couldn't really see much. About three hundred before the loop at the end of the road where I planned to park we finally came out of it and were greeted by some of the most gorgeous scenery Montana has to offer. We were halfway up the hill which seemed like sitting on top of the clouds.

I set up my pocket moxon antenna about twenty feet above the truck with a fiberglass mast and waited for Vince to call on 146.52 to let me know he was in position. After a couple of test beeps we were both hearing each other and it was off to the races. About a minute later BKG was officially in Canada! I set up my FT-891 and got two more QSOs with him for my activation, starting me out with 4 before I ever called CQ. I stayed on the air for eight more QSOs then decided it was time to pack up and begin the journey back to Helena. I submitted our info and Vince soon received his BKG number: #500!

Since that day Vince has jumped no less than fifteen new members in and it is spreading like wildfire amongst our friends up north. BKG has doubled its membership in the last month and shows zero signs of slowing. What an epic adventure. Until the next one, FB VY73 my friends and Dit Dit!

de N7HNT

The Great Montana Shake Out

International Shake Out Day is on October 15th (10/15 at 10:15 a.m. local time), when millions of people worldwide will participate in earthquake drills at work, school, or at home.

At 10:15 a.m. on 10/15, you can join people across Montana in practicing earthquake safety. Local hams can participate in a

directed net exercise shortly after 10:15 on the Belmont repeater: 147.22 MHz (+), 100 Hz tone. CCARC has registered as a group, so you do not need to register individually to take part.

For more information on earthquake safety, and about this event, check out The Great Montana Shake Out website (<https://www.shakeout.org/montana>) or contact Lynn Wagner K7WGR.

Buy / Sell / Trade

Len KC7XH in Helena has five sections of Rohn 20 G, and a rotor with controller and wire. Plus a roof mount antenna mast with rotor bearing. Four tower sections and one top section. The roof top tower is like an A shaped tower with a bearing top that is about five feet tall. He's not looking to get rich but does want some money in exchange.

If interested, the best way to contact him is via text: 406-431-0182.

Calling All Technicians!

Do you have your Technician license and are waiting for the "right" time to upgrade to General?

The Membership Team is considering hosting a General License class this quarter if there is interest in the community.

If you are ready to open up your radio spectrum privileges and really start having fun with HF, email Allen to say you are interested.

membership@ccarc-info.org

Editor: Allen Le Vie KH7AL
Co-Editor: Marla Unruh KM7LIB
Writers: CCARC members

Not yet a member of the CCARC? Join today and help our club & community:
<https://n7rb.org/join-contact>

Capital City Amateur
Radio Club
Helena, Montana

